

GE Healthcare

3.0T Head, Neck, Spine Coil For GE 3.0T MRI Systems Service Manual



Manufactured by:

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GE Healthcare



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Rev. June 13, 2006

LANGUAGE POLICY FOR SERVICE DOCUMENTATION (DOC0620154 REV 2)

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UPOZORENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. - Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. - Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. - Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.
VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. - V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. - Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. - V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanual findes kun på engelsk. - Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. - Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual. - Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for tekniker, operatøren eller patienten.
WAARSCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. - Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. - Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. - Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.
WARNING (EN)	This service manual is available in English only. - If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services. - Do not attempt to service the equipment unless this service manual has been consulted and is understood. - Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles - Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. - Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. - Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. - Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. - Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. - Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	Ce manuel d'installation et de maintenance est disponible uniquement en anglais. - Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire. - Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris. - Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	Diese Serviceanleitung existiert nur in englischer Sprache. - Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen. - Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. - Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.
ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά. - Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης. - Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις. - Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.
FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. - Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése. - Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. - Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatt sérülését eredményezheti.
ADVÖRUN (IS)	Þessi þjónustuhandbók er aðeins fáanleg á ensku. - Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu. - Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilið. - Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. - Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. - Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. - Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

警告 (JA)	このサービスマニュアルには英語版しかありません。
경고 (KO)	- 서비스を担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 - このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 - Σνφ9κ□□φΔ4□□rα ©≠□⊕ΣΥ□□≠r≤∴{B□Δo≤ Y□Λr J□_°□□o}v√vθζκ□□□•Ψ□ }εAB□~Ψγ 본 서비스 매뉴얼은 영어로만 이용할 수 있습니다. - 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다. - 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오. - 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.
BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. - Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. - Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. - Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.
[SPĖJIMAS (LT)	Šis eksploatavimo vadovas yra tik anglų kalba. - Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. - Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo. - Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndbogen finnes bare på engelsk. - Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. - Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. - Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. - Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. - Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. - Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.
AVISO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. - Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.

ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. - Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. - Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. - O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. - Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. - Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. - Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. - Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. - Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. - Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. - Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. - Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. - Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. - Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. - Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. - Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. - Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. - No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. - La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. . - Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. - Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken. - Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
OPOZORILO (SL)	Ta servisni priročnik je na voljo samo v angleškem jeziku. - Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod. - Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli. - Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.
DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. - Eğer müşteri teknisyonu bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. - Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. - Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

1	3T HNS COIL INTRODUCTION	9
1.1	PRODUCT IDENTIFICATION	9
1.2	COMPATIBILITY	9
1.3	SHIPPING CONTENTS	10
1.4	RELATED DOCUMENTATION	12
1.5	ENVIRONMENTAL REQUIREMENTS	12
2	3T HNS COIL INSTALLATION PROCEDURE	13
3	3T HNS COIL MCQA TEST	19
3.1	TOOLS REQUIRED	19
	TOOLS REQUIRED USING LEGACY PHANTOM - TABLE 3-1-1	19
3.2	REQUIRED CONDITIONS	19
3.3	MCQA SETUP WITH LEGACY PHANTOM SET	20
3.4	MCQA SETUP WITH UNIFIED PHANTOM SET	23
3.5	MCQA TEST	27
4	3T HNS COIL TROUBLESHOOTING TIPS	29
4.1	PERSONNEL REQUIREMENTS	29
4.2	OVERVIEW	29
4.3	PRELIMINARY REQUIREMENTS	29
4.4	RECEIVING NO SIGNAL	32
5	3T HNS COIL FIELD REPLACEABLE UNITS	34
5.1	HDXT FORWARD PRODUCTION CONFIGURATION	34
5.2	HDXT UPGRADE CONFIGURATION	34
5.3	COMMON FRU'S	34
5.4	CUSTOMER WEARABLE ACCESSORY	34

ATTENTION!

Installation of this coil **MUST** be performed by a GE field engineer. Please contact your field engineer to arrange for installation.

Attempts to perform installation of this coil without proper system configuration will result in system errors.

1 3T HNS Coil Introduction

1.1 Product Identification

To identify the 3T HNS Coil, refer to the part numbers and descriptions that are provided on the coil rating plates (refer to Table 1).

Table 1: Coil Identification

GEHC part number	5325953-2	5325953-3	5325953-4
Description	3T HNS HNU w/o cable	3T HNS TLU w/o cable	3T HNS NCU

NOTE: Each of the coils (3) subcomponents contains it own label with part numbers and descriptions as shown in Table 1. Also, the HNS HNU part is again comprised of three pieces that all have the same label. These are the Posterior, Anterior, and Horseshoe Adapter.

Figure 1: The 3T HNS Coil (HNU with Posterior and Anterior, NCU, and TLU)



1.2 Compatibility

The 3T HNS Coil is available in two configurations, HDxt Forward Production (FP) and HDxt Upgrade (Upg), which differ only in system connector and cabling. FP configuration is for LPCA's that have only upper A, B, and C ports on the LPCA. Upgrade configuration has upper A and B ports and a lower legacy port.

The 3T HNS coil by Medrad is compatible with HDxt systems only.

1.3 Shipping Contents

1.3.1 Shipping Contents 3T HNS HDxt FP Configuration (HNU with single connector)

Table 2: Box 1 Contents

Part Name	Part Number	Qty.
3T HNS Head Neck Unit – Posterior, Anterior, and Horseshoe Adapter with cable, FP	5325974	1
3T HNS TL Unit with cable	5328289	1
3T HNS Neck Chest Unit	5325953-4	1
3T HNS Operator manual	5325953-10	1
CoilCD	5325975	1

Table 3: Box 2 Contents

Part Name	Part Number	Qty.
3T HNS Head Neck Unit Phantom kit (SiOil) – one cylinder, one brick	5325953-7	1
3T HNS HNU Phantom Positioner	5325953-8	1
MSDS (SiOil), English	5325953-11	2 (one on outside of box)
MSDS (SiOil), Multi-Language	5325953-12	1

Table 4: Box 3 Contents

Part Name	Part Number	Qty.
3T HNS TL Unit Phantom (SiOil) – one large brick phantom	5325953-6	1
MSDS (SiOil), English	5325953-11	2 (one on outside of box)
MSDS (SiOil), Multi-Language	5325953-12	1

Table 5: Box 4 Contents

Part Name	Part Number	Qty.
3T HNS TL Unit Phantom (SiOil) – one large brick phantom	5325953-6	1
MSDS (SiOil), English	5325953-11	2 (one on outside of box)
MSDS (SiOil) , Multi-Language	5325953-12	1

Table 6: Box 5 Contents

Part Name	Part Number	Qty.
3T HNS Pad Kit - one TL pad, one HNU pad, two head pads, and two shoulder pads	5325953-9	1

1.3.2 Shipping Contents 3T HNS HDxt Upg Configuration (HNU with 2 connectors)**Table 7: Box 1 Contents**

Part Name	Part Number	Qty.
3T HNS Head Neck Unit – Posterior, Anterior, and Horseshoe Adapter with cables, Upg	5325957	1
3T HNS TL Unit with cable	5328289	1
3T HNS Neck Chest Unit	5325953-4	1
3T HNS Operator manual	5325953-10	1
CoilCD	5325975	1

Table 8: Box 2 Contents

Part Name	Part Number	Qty.
3T HNS Head Neck Unit Phantom kit (SiOil) – one cylinder, one brick	5325953-7	1
3T HNS HNU Phantom Positioner	5325953-8	1
MSDS (SiOil), English	5325953-11	2 (one on outside of box)
MSDS (SiOil), Multi-Language	5325953-12	1

Table 9: Box 3 Contents

Part Name	Part Number	Qty.
3T HNS TL Unit Phantom (SiOil) – one large brick phantom	5325953-6	1
MSDS (SiOil), English	5325953-11	2 (one on outside of box)
MSDS (SiOil), Multi-Language	5325953-12	1

Table 10: Box 4 Contents

Part Name	Part Number	Qty.
3T HNS TL Unit Phantom (SiOil) – one large brick phantom	5325953-6	1
MSDS (SiOil), English	5325953-11	2 (one on outside of box)
MSDS (SiOil) , Multi-Language	5325953-12	1

Table 11: Box 5 Contents

Part Name	Part Number	Qty.
3T HNS Pad Kit - one TL pad, one HNU pad, two head pads, and two shoulder pads	5325953-9	1

1.4 Related Documentation

- 3T HNS Operator Manual, GE Part Number 5325953-10.

1.5 Environmental Requirements

Storage Requirements

Coil should be stored in the scanner room.

	Transport / Storage	Operating Conditions
Barometric pressure	50 to 106 kPa	70 to 106 kPa
Humidity	5% to 100% non-condensing	10% to 90% non-condensing
Temperature	-40° C to +65° C (-40° F to +149° F)	15° C to 37° C (59° F to 98.6° F)

2 3T HNS Coil Installation Procedure

1. Make sure that the latest software build is loaded on the system prior to installing the HNS coil. The minimum software build is 15.0_M3. If the software build is 15.0_M4A or higher begin installing the coil at step 9. Make sure to read the notes before step 9 before beginning installation on 15.0_M4A software.
2. Open the Config File Manager (CFM) tool from the [Utilities] tab of MR CSD.
3. Go to the Coil config section and:
 - For a Forward Production (FP) system, remove the following 5 HNS configurations:
 - <GE_HNSTLsNoUse>
 - <GE_HNSCTL2>
 - <GE_HNSBrPICsII>
 - <GE_HNSCTBrain2>
 - <GE_HNS NVfullFP>
 - For an Upgrade system, remove the following 6 HNS configurations:
 - <GE_HNSTLsNoUse>
 - <GE_HNSCTBrain2>
 - <GE_HNSBPCsII4>
 - <GE_HNSBPCsII3>
 - <GE_HNSCsICTs0>
 - <GE_HNSBPCsII1>

Remove the HNS configurations by doing the following. See Figure 2.

- a. Select the coil from the coil name
- b. Then select **Remove Coil** in the lower right side of the GUI
- c. Select **Yes**
- d. Repeat for the 5 mentioned coils for FP system, or for the 6 mentioned coils for the upgrade system
- e. Select **Save Changes**, then select **Yes**
- f. Select **Quit** to exit Config File Manager.

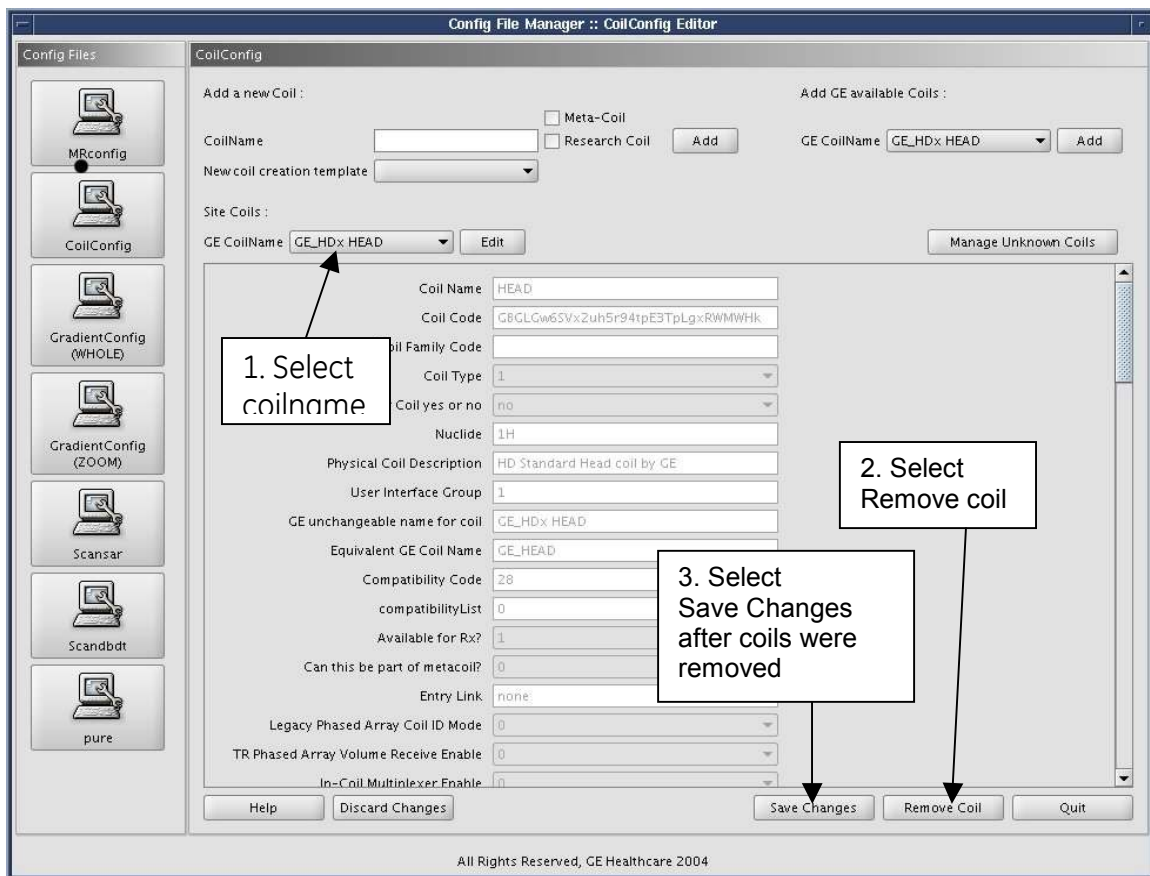


Figure 2: Removing coil config

4. Put the CoilCD into External / Internal DVD Drive.
5. Open a command window and type the following commands:


```
su
< root password >
mount /mnt/cdrom
/mnt/cdrom/coil_prescan_update
```
6. Launch CSD, go to [Utilities], Select [Install Coil CD].
7. Select [Click here to start this tool], an xterm window will open.

If the CD has already been installed on the system you will get a message that states the packages are already on the system disk. Eject CD and proceed to step 8.

It may take up to 10 minutes for window to open and for CD to be ejected automatically based on system configuration. A message in the xterm window should appear stating "Coil Media Installation Successful". At this time the CD is ejected. Remove it then close the xterm window.

8. Reboot the System and login as SDC.

Note: Each physical configuration of the HNS coil must be installed separately for the system to recognize it. **Connecting all the coil components together will not result in the coil getting completely configured.** Setups 1 through 4 show the valid physical coil configurations for installation.

Note: Ensure that the coil components have a snug fit with each other.

If the coil could not be configured successfully, **DO NOT ATTEMPT TO INSTALL THE COIL MANUALLY WITH CONFIG FILE MANAGER OR VIA ANY OTHER METHOD**, please call the online center for additional information.

9. Connect the coil to the LPCA as shown in Setup 1. A pop-up will appear on the monitor after connecting the coil stating “ A new coil has been connected to the system, Do you want to configure?” Select **YES**. Then select **OK**.
10. After proper configuration of setup 1, remove the coil cable from LPCA, prior to changing coil configuration for setup 2.
11. Repeat the steps 9 and 10 for setups 2 through 4.

- **Setup 1:** Head Neck Unit only (Posterior + Anterior)



Note: The pictures show Head Neck Unit in FP configuration – single connector. Connect cable to B-port.
The Upgrade configuration will have an additional cable that is connected to the legacy port.

- **Setup 2:** Head Neck Unit only (Posterior + Horseshoe Adapter)



- **Setup 3:** Posterior + Anterior + TL + NCU



Note: Connect cable from head section to B-Port, and TL section to A-Port. Shown in the pictures are the FP configurations; the Upgrade configuration will have an additional cable that is connected to the legacy port.

- **Setup 4:** Posterior + Horseshoe Adapter + TL + NCU



12. After all four coil configurations have been entered, reboot the system.

13. Prescribe new exam. Enter the patient information and click on the coil selection button. Verify that the HNS coil is present when clicking on the Currently Connected button under Coil Type on the scanner user interface. For example, if you use Setup 1 you will see the clinical coil modes as shown in Figure 3.

In forward production systems, the presence of coil entries like those shown is a confirmation that the coil installed properly. If you do not see the HNS coil modes, **DO NOT ATTEMPT TO INSTALL THE COIL MANUALLY**, call the online center for additional information.

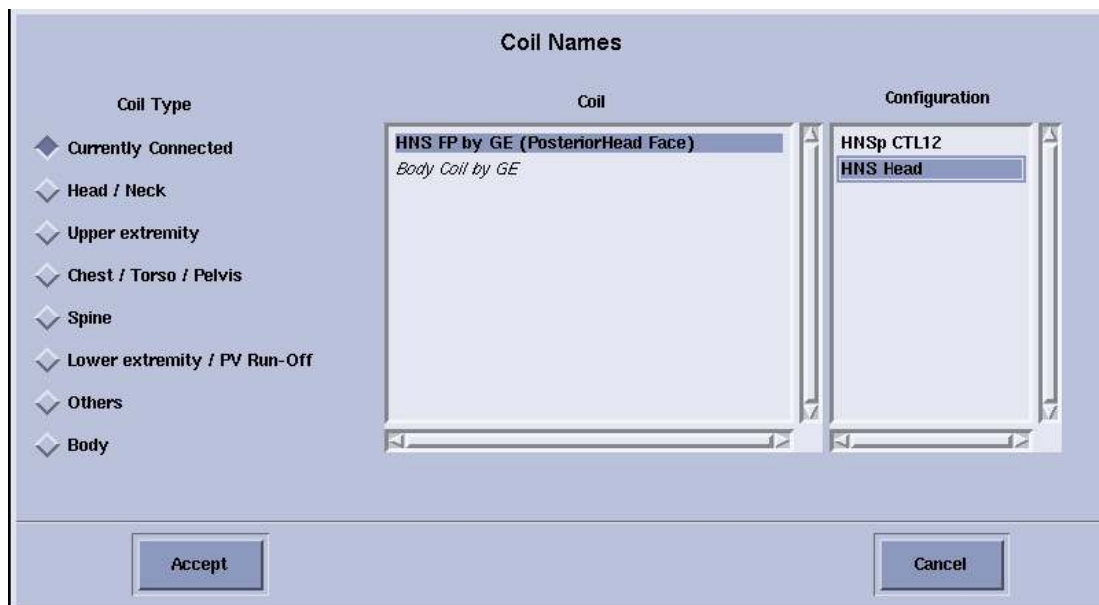


Figure 3: Confirmation of proper configuration

14. Perform MCQA test on coil.

Reminder: Complete Product Locator information for serialized components. Failure to do this may result in failure of your site to receive future FMI's.

3 3T HNS Coil MCQA Test

Follow this procedure to perform the MCQA test. Either the legacy phantom kit (shipped with coil) or the unified phantom kit (shipped with system) can be used for MCQA.

NOTE:

- **The coils must be tested in 3 configurations:**
 - 1. The TL unit only**
 - 2. Posterior + NCU + Horseshoe Adapter + TL**
 - 3. Posterior + Anterior**

This is done to ensure modularity in design and in diagnostics.

- **The MCQA tool can ONLY run on the above mentioned individual coil configurations and NOT all coils configurations at the same time. MCQA will not run and will report that the configuration is invalid if an attempt is made to scan with an unsupported combination of coils.**

3.1 TOOLS REQUIRED

TOOLS REQUIRED USING LEGACY PHANTOM - TABLE 3-1-1

Description	GE Part #	Medrad Part #
3T HNS Head Neck Unit Phantom kit (SiOil) – one cylinder, one brick, qty. 1 ea	5325953-7	3017226-1 and 3017226-2
3T HNS TL Unit Phantom (SiOil) – one large brick phantom, qty. 2	5325953-6	3017290
3T HNS HNU phantom positioner	5325953-8	N/a

TOOLS REQUIRED USING UNIFIED PHANTOM - TABLE 3-1-2

Description	GE Part #	Qty
Large Cylindrical Unified Phantom, SiOil	5342679-2	1
TL Unified Phantom, SiOil	5343347-2	2
Phantom positioner, HNS	5344844	1

3.2 Required Conditions

The appropriate coil configuration files must be installed to run this tool. See section 4.3.3.

3.3 MCQA setup with Legacy Phantom Set

1. Record the coil serial #s of the following coils:



Figure 4

HNU Serial # (The serial #s of Posterior, Anterior, and Horseshoe Adapter should all match)

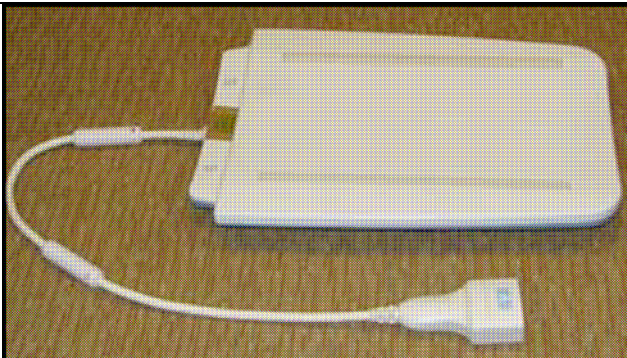


Figure 5

TL Serial #

2. To position the coil on the patient cradle refer to Figure 6 - Figure 9
3. Remove all pads before placing the phantoms in the coil.



Figure 6: TL Unit and phantom setup



Figure 7: Posterior + NCU + Horseshoe Adapter + TL coil and phantom setup, step 1 (handle of brick phantom faces downward)



Figure 8: Posterior + NCU + Horseshoe Adapter + TL coil and phantom setup, step 2



Figure 9: Posterior + Anterior coil and phantom setup

4. Connect the appropriate cables to the LPCA (Example of the Upgrade Configuration Posterior + NCU + Horseshoe Adapter + TL shown in Figure 10, other configurations may have different connections):



Figure 10: Connecting the coil

5. Landmark as shown in Figure 11 - Figure 13 as the case may be and advance to scan .
Note: For the TL only setup, the handle of the middle phantom must align with the digit 5 and the landmark line must pass through the handle/digit 5 as shown in Figure 11.



Figure 11: TL Only Setup Landmark



Figure 12: Posterior + NCU + Horseshoe Adapter + TL Setup Landmark (on the crosshairs of the NCU)



Figure 13: Posterior + Anterior Setup Landmark (at inferior edge of arch)

3.4 MCQA setup with Unified Phantom Set

3.4.1 Coil-Phantom Setup Procedure HNU Posterior and Anterior sections

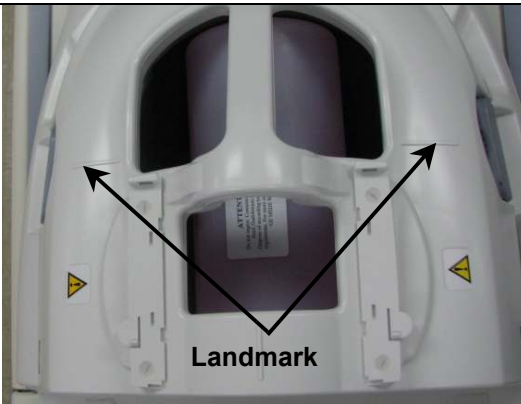
Visual	Description
 Figure 14	Place HNU Posterior section on the table as shown in Figure 14

**Figure 15**

Place HNS positioner and then place large cylindrical unified phantom in the positioner as shown in Figure 15.

**Figure 16**

Carefully align the HNU anterior component with the connector pins over the HNU Posterior section and latch both the coil sections into place as shown in Figure 16. (The scanner will not operate if the coil sections are not latched correctly).

**Figure 17**

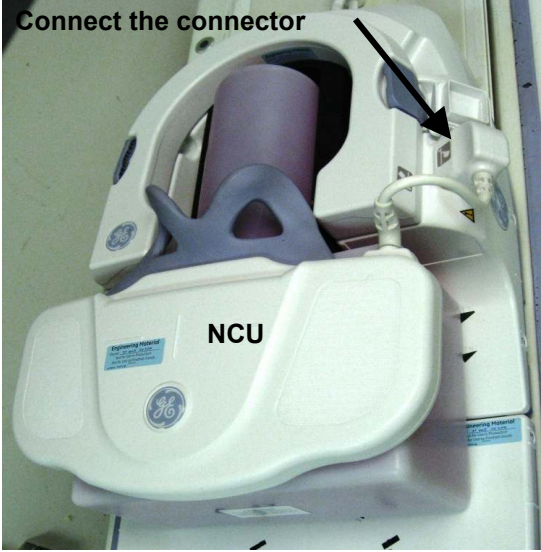
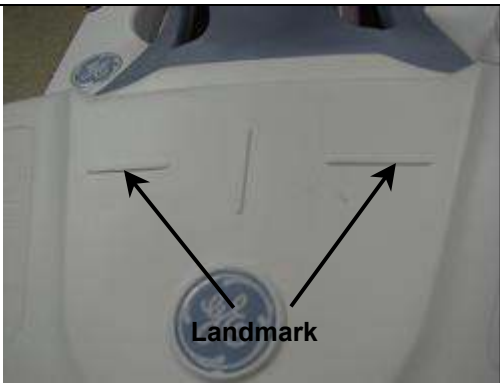
Connect the coil to respective port of the system LPCA.

Landmark the coil on the markings present on the face component indicated through arrows as shown in Figure 17 and press advance to scan.

Refer to Section 3.5 of this document to run the MCQA Tool with the Unified Phantom Set.

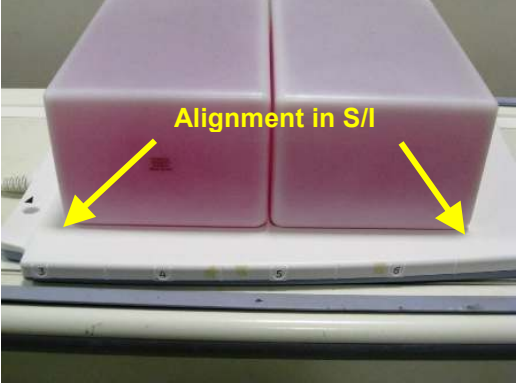
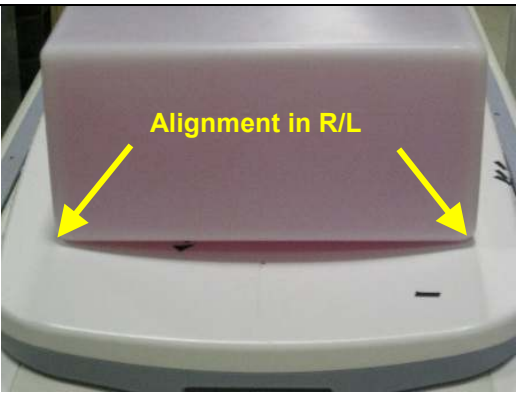
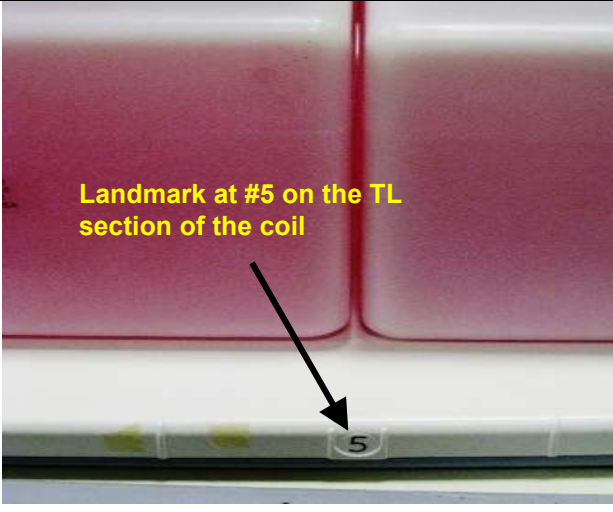
3.4.2 Coil-Phantom Setup Procedure HNU Posterior, Horseshoe Adapter, NCU and TL section

Visual	Description
 Figure 18	Place the HNU Posterior section, TL section and Horseshoe adapter on the patient table. Carefully latch the Horseshoe adapter into the connectors of the coil Figure 18 Place the phantom positioner as shown in the figure.
 Figure 19	Place the Large cylindrical unified phantom as shown in Figure 19.
 Figure 20	Place the TL unified phantom over the junction of the TL section and head components as shown in Figure 20. Ensure that TL unified phantom is in contact with cylindrical unified phantom as shown in Figure 20. The cylindrical phantom will have to be moved forward in the positioner.

 Connect the connector Figure 21	Connect the neck, chest unit (NCU) to the HNU posterior section as shown in Figure 21. The complete setup is shown in Figure 21.
 Figure 22	Landmark on the markings over the NCU section of the coil as indicated in Figure 22 Press advance to scan. Refer to Section 3.5 of this document to run the MCQA Tool with the Unified Phantom Set.

3.4.3 Coil-Phantom Setup Procedure TL section

Visual	Description
 Figure 23	TL section Place the TL section of HNS coil on the table as shown in Figure 23.

 Figure 24	Place TL unified phantoms on the TL section of the coil and align them properly in S/I direction as shown in figure Figure 24.
 Figure 25	Align TL unified phantoms properly in R/L direction of the TL section of the coil as shown in Figure 25.
 Figure 26	Connect the coil to respective port of system LPCA. Landmark on #5 present on the side of the TL section of the coil as shown in Figure 26 and press advance to scan. Refer to Section 3.5 of this document to run the MCQA Tool with the Unified Phantom Set.

3.5 MCQA Test

1. At the operator workstation, end the previous exam (if required).

2. From the Common Service Desktop, select the [Image Quality] tab, choose “Multi-Coil QA Tool”, and then “Click here to start this tool” as shown in Figure 27.

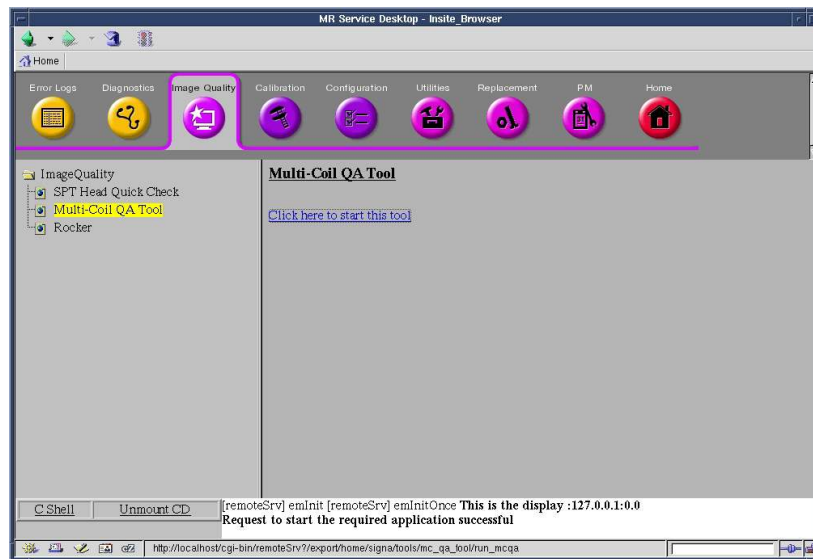


Figure 27: MCQA from the Service Desktop

3. For further instructions on the MCQA tool, it can be found on the Signa HDxt Service Methods CD or the web at: [Troubleshooting -> Image Quality -> Multi-Coil Quality Assurance Tool](#).

4 3T HNS Coil Troubleshooting Tips

4.1 Personnel Requirements

The following procedure requires one person for about 30 minutes.

4.2 Overview

The following tips can be used to troubleshoot common problems with the 3T HNS coil.

4.3 Preliminary Requirements

4.3.1 *Tools and Test Equipment*

Some tests may require Phantoms for IQ Testing.
Digital Multi Meter.

4.3.2 *Replacement Parts*

See Section 5: 3T HNS Coil Field Replaceable Units

4.3.3 *Required Conditions*

All different configurations of the coil for the different setups must be installed to run the MCQA tool. The following table shows all the different configurations and their corresponding coil setups.

Components Connected	Configurations Available
Posterior and Anterior	HNSp CTL12
	HNS HEAD

Components Connected	Configurations Available
Posterior and Adapter	HNS CTL12

Components Connected	Configurations Available
Posterior, Adapter, and TL	HNS CTLTOP
	HNS CTLMID
	HNS CTLBOT
	HNS CTL123
	HNS CTL234
	HNS CTL345
	HNS CTL456
	HNS CTL12
	HNS CTL23
	HNS CTL34
	HNS CTL45

	HNS CTL56
Components Connected	Configurations Available
Posterior, Anterior, and TL	HNS HEAD
	HNS 35FOV
	HNS 48FOV
	HNSp CTLTOP
	HNSp CTLMID
	HNS CTLBOT
	HNSp CTL123
	HNSp CTL234
	HNS CTL345
	HNS CTL456
	HNSp CTL12
	HNSp CTL23
	HNS CTL34
	HNS CTL45
	HNS CTL56

Components Connected	Configurations Available
Posterior, Anterior, TL and NCU	HNS Neck
	HNS NeckChest
	HNS NVFull
	HNS NVHead
	HNS BrachPlx
	HNS- CTLTOP
	HNS- CTLMID
	HNS CTLBOT
	HNS- CTL123
	HNS- CTL234
	HNS CTL345
	HNS CTL456
	HNS- CTL12
	HNS- CTL23
	HNS CTL34
	HNS CTL45

	HNS CTL56
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Components Connected	Configurations Available
Posterior, Adapter, TL, and NCU	HNS BrachPlex
	HNS_ CTLTOP
	HNS_ CTLMID
	HNS CTLBOT
	HNS_ CTL123
	HNS_ CTL234
	HNS CTL345
	HNS CTL456
	HNS_ CTL12
	HNS_ CTL23
	HNS CTL34
	HNS CTL45
	HNS CTL56

Components Connected	Configurations Available
Posterior, Anterior and NCU	HNS Neck
	HNS NeckChest
	HNS NVFull
	HNS NVHead
	HNS- CTL12

Components Connected	Configurations Available
TL only	HNS CTLBOT
	HNS CTL345
	HNS CTL456
	HNS CTL34
	HNS CTL45
	HNS CTL56

4.4 Receiving No Signal

4.4.1 Receiving No Signal

Problem:

You are unable to pre-scan or are scanning, yet receiving no signal, or receiving partial signal from coil elements.

Possible Solution:

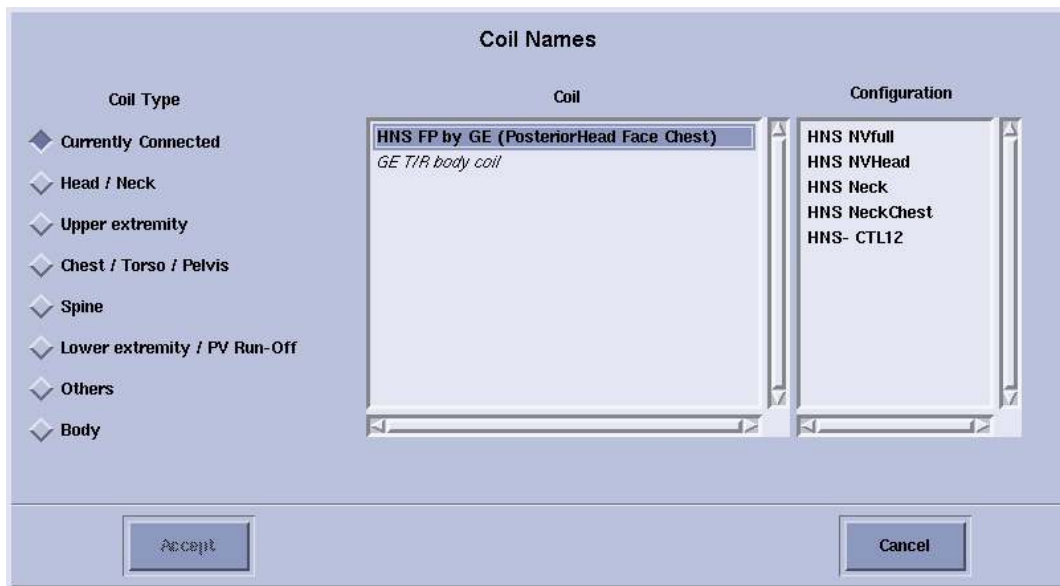
Verify the green light above the port (Port A or B, Port A and B, depends on which port is used to plug in the coil) is illuminated. This indicates the coil is properly plugged into the system and the coil I.D. on the coil is properly installed.

NOTE: For Forward Production systems, use Port A (Hypertronics, 8 channels) for TL Unit (Thoracic Lumbar), and Port B (Hypertronics, 16 Channel) for Head Neck Unit.

- If the TL is used by itself, then it can also be connected to port B.
- For upgrade systems: use Port A (Hypertronics, 8 channels) for TL (Thoracic Lumbar) Unit, Port B (Hypertronics, 8 Channel) and 8 channel legacy 3T connector for Head Neck Unit.

Verify that the system has correctly detected the coil by checking in the “Currently Connected” in the GUI to select coils in Rx as shown in Figure 28.

Figure 28: “Currently Connected” Coils Window In The Scan Screen



- Verify that the landmark is correct and that the cradle (patient bed) has not unlatched.
- Verify that the scan locations and any FOV offsets are correct.
- Perform a continuity check on the output cable (Section 5.4.4) if you lose some signals from some of coil channels. The continuity check must be performed by a GE authorized Service Engineer.
- Verify that the coil is positioned with the cable exiting towards the bore.

If you still cannot get a signal, try to scan (transmit and receive) with the body coil. For this test, be sure to remove the imaging coil from the magnet bore before you scan with the body coil. If you still receive no signal the problem probably lies with the MR system. If the scan completes successfully, there is probably a problem with the HNS coil.

You may also try to scan with a similar type (receive only) coil. If you are unable to scan with this coil, there may be a system problem related to this particular coil type.

4.4.2 Image Quality

Problem:

Poor IQ, shaded images, or if MCQA fails

Possible Solution:

- Perform an output cable check (Section 4.4.4).
- Verify that there are no loops in the cables.
- Verify that there are no metal or ferromagnetic objects close to the coil, patient or magnet (i.e., safety pin, hair pin).
- Verify that the coil is properly positioned.
- Verify that your center frequency is within the frequency adjustment range for your system.
- Verify that the R1, R2 and TG values from the pre-scan are within normally expected ranges.

If you have not done so already, perform a system Quality Assurance MCQA test. If the values you obtain do not fall within normal operating parameters, investigate this further by performing a phantom scan with the body coil. For this test, be sure to remove the imaging coil from the magnet bore before you scan with the body coil. If you still have the same problems, there is probably an MR system problem. If the body coil scan is satisfactory, acquire a scan using both another coil of the same type (receive-only, phased array or transmit/receive, whichever applies) and the same system coil selection. If the image quality is visibly improved, there may be a problem with the Head Neck Spine coil. Contact GE for further assistance. If the image quality still suffers, there may be a system problem related to imaging with this type of coil.

4.4.3 Artifacts

Problem:

There is a black line, strip lines, or signal void on the image.

Possible Solution:

- Verify that there is no metal present in the area being scanned in, or on the patient.
- Verify that no hair is trapped in the connectors between the anterior and posterior sections of the head unit. This may appear as intermittent oscillations.
- Verify the cable connector of the NCU is fully engaged into the receptacle on the posterior section of the head unit. If the connector is not fully engaged, there may be a coil not present error or there may be intermittent oscillations.
- If you have oscillation of the preamp you will see striped lines.

4.4.4 Output Cable Check

Visually inspect all coaxial connectors and pins on the system plug. If there are any broken, deformed or recessed pins or coaxial connectors, replace the coil component with the damaged cable. Reference section 5 for FRU part numbers.

5 3T HNS Coil Field Replaceable Units

5.1 HDxt Forward Production Configuration

Table 12: 3T HNS Forward Production Configuration FRUs

Part Number	Description
5325974 (consisting of 5325953-2 and 5325974-2)	3T Head Neck Unit - Posterior, Anterior, Horseshoe adapter FP (with cable)
5325953-4	3T HNS Neck Chest Unit
5328289 (consisting of 5325953-3 and 5325953-5)	3T HNS TL Unit (with cable)

5.2 HDxt Upgrade Configuration

Table 13: 3T HNS Upgrade Configuration FRUs

Part Number	Description
5325957 (consisting of 5325953-2, 5325957-2, and 5325957-3)	3T Head Neck Unit - Posterior, Anterior, Horseshoe adapter UG (with cables)
5325953-4	3T HNS Neck Chest Unit
5328289 (consisting of 5325953-3 and 5325953-5)	3T HNS TL Unit with cable

5.3 Common FRU's

Table 14: 3T HNS Common FRUs to both configurations

Part Number	Description
GEHC: 5325953-6 Medrad: 3017290	Legacy 3T HNS TL Unit Phantom (one large brick SiOil Phantom)
GEHC: 5325953-7 Medrad: 3017226-1 and 3017226-2	Legacy 3T HNS Head Neck Unit Phantom Kit (one cylinder and one brick shaped SiOil phantom)
5325953-8	Legacy 3T HNS Head Neck Unit Phantom Positioner
5325953-14	3T HNS Mirror
5342679-2	Unified Large Cylindrical Unified Phantom, SiOil
5343347-2	Unified TL Unified Phantom, SiOil
5344844	Unified Phantom positioner, HNS

5.4 Customer Wearable Accessory

Table 15: Pads

Part Number	Description
GEHC: 2418081 or E8801WG	Pad Kit (consists of Wedge Pad, Transition Pad, Left Shoulder Pad, Right Shoulder Pad, Main T-L Pad, Main Patient Pad, Head Support Pad)

Revision History

Medrad Document Revision History

Rev	Date	Author	Primary Reason For Change
A	13JAN2009	Julia Klinge	Initial Release
B	31MAR2009	Rebecca David	- Added upgrade configurations, added step to update the prescan_config.cfg file, and reminder to follow product locator process to section 2 - Section 4.6: Tables 16-19 updated with correct cable mapping. - Section 6: Changed CoilID removal to be after cable removal and added step 14 to section 6.4
C	05MAY2009	Rebecca David	- Cover sheet: Added Medrad information - Added multi-language warning statements to beginning of document - Section 2, step 3: removed <GE_HNSCslCTs1> and replaced with <GE_HNSBPCsl1> coil config. - Section 2, step 3: removed the command for unmounting the CD as it is still needed in the drive to install. Added text to step 7 as to what will display if CD has been installed on the system. - Section 4.5.1.1: Removed upper images on figure 19 and inserted new cable map image for cable B; added reference on where to find cable removal instructions. - Sections 4.6.1.1 and 4.6.1.2: added reference to figure 19. - Section 4.7: Removed table for different jumper configuration between FP and Upgrade. - Sections 3.1 and 5.1: Added cross-reference to Medrad phantom part numbers. - Section 6.4: Inserted new image for figure 34 and figure 35 - Section 6.4: Added how to connect S1 and S2 cables to Coil ID board. - Section 6.4: Added image for routing RF cables after new cable installation.
D	12JUNE2009	Rebecca David	- Section 4.4: Removed reference to replace the cable if it is damaged and said to replace coil component. - Section 4.5: RF Signal Channels Check – removed - Section 4.6: DC Lines Continuity Check – removed - Section 4.7: Configuration of Jumpers on Interface Board – removed - Section 5.1: Removed cables from table 20 - Section 5.2: Removed cables from table 21 - Section 6: 3T HNS Coil Cable Replacement - removed
E	01DEC2009	Rebecca David	- Added damage in transportation instructions - Updated language policy - Added information on unified phantom set-up (section 3.4) and FRUs (section 5)
F	30SEP2010	Rebecca David	- Updated language policy to rev 2 - Added information on Accessory Pads, section 5.4